Effective Area of a Receiving Antenna

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Lecture Objective

By the end of this lecture, you will be able to:

- Understand the concept of effective area,
- Evaluate the effective area of a specified antenna given its gain.

Antennas in Transmit and Receive

- In the previous lectures, we considered antennas to be directional radiators of electromagnetic energy. In the direction of maximum radiation, in the far field, the electromagnetic energy could be modelled locally as a plane wave.
- Now, we consider how a receiving antenna captures some of the energy in an incident plane and delivers it to a load.

Effective Area of Receiving Antenna

- The ***effective area*** (a.k.a. effective aperture or receiving cross section) of an antenna is given by

$$A_e = \frac{P_{\text{int}}}{S_{\text{i}}}$$

intercepted power delivered to load (W)

effective area (m^2)

power density of incident wave (W/m^2)

Effective Area and Directivity

- The effective area of a receive antenna is related to its gain by

$$A_e = \frac{\lambda^2 G}{4\pi}$$

- For a lossless and matched antenna, the gain is equal to the directivity and we have

$$A_e = \frac{\lambda^2 D}{4\pi}$$

- Strictly speaking, effective area depends on the direction of arrival of the incident wave, but we usually assume that the antenna is oriented so as to maximize the received power.

Question

- What is the effective area of a Hertzian dipole?

Effective Area of Wire and Aperture Antennas

- The effective area of a wire antenna can be many times larger than its physical area.
- The effective area of an aperture antenna will always be smaller than its physical area.
- The aperture efficiency of an aperture antenna can be defined as

$$\eta_{ap} = \frac{A_e}{A_p}$$

Physical area of aperture

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